

Se instala WSL con ubuntu

```
Static hostname: SAULI
Icon name: computer-container
Chassis: container
Machine ID: 3597858406af440f9dc3ba8fbc2c1abb
Boot ID: 018dd29145b4494d9a1298ead2acd13b
Virtualization: wsl
Operating System: Ubuntu 26.04 LTS
Kernel: Linux 6.18.33.2-microsoft-standard-WSL2
Architecture: x86_64
saula@SAULI:~$
```

Open router:

```
saula@SAULI:~$ export OPENROUTER_API_KEY="sk-or-v1-f891fe5cf0df57d66d5bafcb4fe37b938eefb0215f743955e381f45ffdd151"
saula@SAULI:~$ echo "$OPENROUTER_API_KEY"
sk-or-v1-f891fe5cf0df57d66d5bafcb4fe37b938eefb0215f743955e381f45ffdd151
```

2.3

```
saula@SAULI:~$ curl https://openrouter.ai/api/v1/chat/completions \
-H "Content-Type: application/json" \
-H "Authorization: Bearer $OPENROUTER_API_KEY" \
-d '{
  "model": "openrouter/free",
  "messages": [
    {
      "role": "user",
      "content": "Explicue en una oración qué es un ticket de soporte técnico."
    }
  ]
}'

{"id":"gen-1784771232-crESWflevWbrcsPn4y","object":"chat.completion","created":1784771232,"model":"nvidia/nemotron-3-nano-30b-a3b-free","provider":"Nvidia","system_fingerprint":null,"service_tier":null,"choices":[{"index":0,"logprobs":null,"finish_reason":"stop","native_finish_reason":"stop","message":{"role":"assistant","content":"Un ticket de soporte técnico es un registro formal que documenta una solicitud, problema o incidencia de un usuario, asignando un identificador único para que el equipo de soporte pueda rastrear, priorizar y resolverla de manera eficiente.", "refusal":null,"reasoning":"We need to respond in Spanish, one sentence, explaining what a technical support ticket is. Should be concise, one sentence.", "reasoning_details":{"type":"reasoning.text","text":"We need to respond in Spanish, one sentence, explaining what a technical support ticket is. Should be concise, one sentence.", "format":"unknown","index":0}}}], "usage":{"prompt_tokens":32,"completion_tokens":81,"total_tokens":113,"cost":0,"is_byok":false,"prompt_tokens_details":{"cached_tokens":0,"cache_write_tokens":0,"audio_tokens":0,"video_tokens":0},"cost_details":{"upstream_inference_cost":0,"upstream_inference_prompt_cost":0,"upstream_inference_completions_cost":0},"completion_tokens_details":{"reasoning_tokens":32,"image_tokens":0}}saula@SAULI:~$ curl https://openrouter.ai/api/v1/chat/completions \
-H "Content-Type: application/json" \
-H "Authorization: Bearer $OPENROUTER_API_KEY" \
-d '{
  "model": "openrouter/free",
  "messages": [
    {
      "role": "user",
      "content": "Explicue en una oración qué es un ticket de soporte técnico."
    }
  ]
}'
```

2.4

```
{
  "id": "gen-1784771261-1hvYA0ort02uAE3x0Bdh",
  "object": "chat.completion",
  "created": 1784771261,
  "model": "nvidia/nemotron-nano-9b-v2:free",
  "provider": "Nvidia",
  "system_fingerprint": null,
  "service_tier": null,
  "choices": [
    {
      "index": 0,
      "logprobs": null,
      "finish_reason": "stop",
      "native_finish_reason": "stop",
      "message": {
        "role": "assistant",
        "content": "\n\nHPC\n\n",
        "refusal": null,
        "reasoning": "Okay, let's tackle this ticket classification. The user mentioned that the job sent to Slurm is pending due to a QoS restriction. First, I need to recall what each category refers to.\n\nHPC (High-Performance Computing) usually involves clusters and job scheduling. Slurm is a common workload manager for HPC environments. QoS (Quality of Service) restrictions are often related to resource allocation in HPC systems.\n\nSSO (Single Sign-On) is about authentication, which doesn't seem relevant here. Plataforma web (Web Platform) would involve web applications or servers. Balanceador (Load Balancer) deals with distributing traffic. Base de datos (Database) is about data storage. Redes (Networks) would involve network infrastructure issues.\n\nThe ticket is about a job stuck in Slurm because of QoS. Since Slurm is used in HPC for job scheduling and QoS policies are part of resource management in HPC, this fits under HPC. The other categories don't relate to job scheduling or QoS in a computing cluster context. So the correct category should be HPC.\n\n",
        "reasoning_details": {
          "type": "reasoning.text",
          "text": "Okay, let's tackle this ticket classification. The user mentioned that the job sent to Slurm is pending due to a QoS restriction. First, I need to recall what each category refers to.\n\nHPC (High-Performance Computing) usually involves clusters and job scheduling. Slurm is a common workload manager for HPC environments. QoS (Quality of Service) restrictions are often related to resource allocation in HPC systems.\n\nSSO (Single Sign-On) is about authentication, which doesn't seem relevant here. Plataforma web (Web Platform) would involve web applications or servers. Balanceador (Load Balancer) deals with distributing traffic. Base de datos (Database) is about data storage. Redes (Networks) would involve network infrastructure issues.\n\nThe ticket is about a job stuck in Slurm because of QoS. Since Slurm is used in HPC for job scheduling and QoS policies are part of resource management in HPC, this fits under HPC. The other categories don't relate to job scheduling or QoS in a computing cluster context. So the correct category should be HPC.\n\n",
          "format": "unknown",
          "index": 0
        }
      },
      "usage": {
        "prompt_tokens": 89,
        "completion_tokens": 238,
        "total_tokens": 327,
        "cost": 0,
        "is_byok": false,
        "prompt_tokens_details": {
          "cached_tokens": 0,
          "cache_write_tokens": 0,
          "audio_tokens": 0,
          "video_tokens": 0
        },
        "cost_details": {
          "upstream_inference_cost": 0,
          "upstream_inference_prompt_cost": 0,
          "upstream_inference_completions_cost": 0,
          "completion_tokens_details": {
            "reasoning_tokens": 266,
            "image_tokens": 0
          }
        }
      }
    }
  ]
}
```

2.5

```
{
  "id": "gen-1784771373-RuavvmVmgjARTLNUJbX",
  "object": "chat.completion",
  "created": 1784771373,
  "model": "nvidia/nemotron-3-super-120b-a12b:free",
  "provider": "Nvidia",
  "system_fingerprint": null,
  "service_tier": null,
  "choices": [
    {
      "index": 0,
      "logprobs": null,
      "finish_reason": "stop",
      "native_finish_reason": "stop",
      "message": {
        "role": "assistant",
        "content": "\n\nSSO\n\n",
        "refusal": null,
        "reasoning": "We need to output JSON with fields 'servicio' and 'confianza'. The ticket: 'No puedo iniciar sesión mediante Keycloak en la aplicación institucional.' Keycloak is an identity and access management solution, often used for SSO. So service: SSO. Confidence high, maybe 0.95. Provide JSON.\n\n",
        "reasoning_details": {
          "type": "reasoning.text",
          "text": "We need to output JSON with fields 'servicio' and 'confianza'. The ticket: 'No puedo iniciar sesión mediante Keycloak en la aplicación institucional.' Keycloak is an identity and access management solution, often used for SSO. So service: SSO. Confidence high, maybe 0.95. Provide JSON.\n\n",
          "format": "unknown",
          "index": 0
        }
      },
      "usage": {
        "prompt_tokens": 93,
        "completion_tokens": 95,
        "total_tokens": 188,
        "cost": 0,
        "is_byok": false,
        "prompt_tokens_details": {
          "cached_tokens": 0,
          "cache_write_tokens": 0,
          "audio_tokens": 0,
          "video_tokens": 0
        },
        "cost_details": {
          "upstream_inference_cost": 0,
          "upstream_inference_prompt_cost": 0,
          "upstream_inference_completions_cost": 0,
          "completion_tokens_details": {
            "reasoning_tokens": 73,
            "image_tokens": 0,
            "audio_tokens": 0
          }
        }
      }
    }
  ]
}
```

3.2

```

root@SAULI:~# mkdir -p ~/clasificador_tiquetes_openrouter
root@SAULI:~# cd ~/clasificador_tiquetes_openrouter
root@SAULI:~/clasificador_tiquetes_openrouter# python3 -m venv .venv
root@SAULI:~/clasificador_tiquetes_openrouter# source .venv/bin/activate
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter#
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# pip install --upgrade pip
Requirement already satisfied: pip in ./venv/lib/python3.14/site-packages (25.1.1)
Collecting pip
  Using cached pip-26.1.2-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.1.2-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.1.1
    Uninstalling pip-25.1.1:
      Successfully uninstalled pip-25.1.1
Successfully installed pip-26.1.2
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# |

```

3.3

3.4

```

postgres@SAULI:~$ psql
psql (18.4 (Ubuntu 18.4-0ubuntu0.26.04.1))
Type "help" for help.

postgres=# CREATE TABLE IF NOT EXISTS public.servicio (
    id BIGSERIAL PRIMARY KEY,
    nombre VARCHAR(150) NOT NULL UNIQUE,
    descripcion TEXT,
    activo BOOLEAN NOT NULL DEFAULT TRUE
);
CREATE TABLE
postgres=# INSERT INTO public.servicio (nombre, descripcion)
VALUES
    ('HPC', 'Clúster, Slurm, colas, nodos, GPU, software científico y almacenamiento HPC'),
    ('SSO', 'Keycloak, SAML, OpenID Connect, LDAP, autenticación y atributos'),
    ('Plataforma web', 'Drupal, WordPress, Joomla, Apache, PHP-FPM, Varnish y hospedaje web'),
    ('Balanceador', 'HAProxy, Traefik, balanceo de carga, VIP, certificados y proxy inverso'),
    ('Base de datos', 'PostgreSQL, MariaDB, consultas SQL, conexiones y respaldos'),
    ('Redes', 'DNS, VPN, firewall, conectividad, puertos y enrutamiento'),
    ('Otro', 'Solicitudes que no corresponden a los servicios anteriores')
ON CONFLICT (nombre) DO UPDATE
SET descripcion = EXCLUDED.descripcion,
    activo = TRUE;
INSERT 0 7
postgres=# |

```

3.5 y 3.6

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python crm_mcp_server.py
```



FastMCP 3.4.4
https://gofastmcp.com

🖨️ Server: CRM PostgreSQL, 3.4.4
🚀 Deploy free: https://horizon.prefect.io

```
[07/22/26 20:33:34] INFO      Starting MCP server 'CRM PostgreSQL' with transport 'http' on http://127.0.0.1:8000/mcp transport.py:361
INFO:      Started server process [5210]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

3.7

```
saula@SAULI: $ ss -lntp | grep 8000
LISTEN 0      2048        127.0.0.1:8000      0.0.0.0:*
saula@SAULI: $ ls
clasificador_tiquetes_openrouter  probar_mcp.py
saula@SAULI: $ cd clasificador_tiquetes_openrouter/
saula@SAULI:~/clasificador_tiquetes_openrouter$ source .venv/bin/activate
-bash: .venv/bin/activate: No such file or directory
saula@SAULI:~/clasificador_tiquetes_openrouter$ source .venv/bin/activate
-bash: .venv/bin/activate: No such file or directory
saula@SAULI:~/clasificador_tiquetes_openrouter$ sudo -i
[sudo: authenticate] Password:
})root@SAULI:~# cd clasificador_tiquetes_openrouter/
root@SAULI:~/clasificador_tiquetes_openrouter# source .venv/bin/activate
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# nano probar_mcp.py
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python probar_mcp.py
Conexión MCP correcta
Herramientas disponibles:
- listar_servicios
- obtener_servicio_por_id
- buscar_servicios

Servicios devueltos por PostgreSQL:
CallToolResult(content=[TextContent(type='text', text='[{"id":4,"nombre":"Balanceador","descripcion":"HAProxy, Traefik, balanceo de carga, VIP, certificados y proxy inverso"}, {"id":5,"nombre":"Base de datos","descripcion":"PostgreSQL, MariaDB, consultas SQL, conexiones y respaldos"}, {"id":1,"nombre":"HPC","descripcion":"Clúster, Slurm, colas, nodos, GPU, software científico y almacenamiento HPC"}, {"id":7,"nombre":"Otro","descripcion":"Solicitudes que no corresponden a los servicios anteriores"}, {"id":3,"nombre":"Plataforma web","descripcion":"Drupal, WordPress, Joomla, Apache, PHP-FPM, Varnish y hospedaje web"}, {"id":6,"nombre":"Redes","descripcion":"DNS, VPN, firewall, conectividad, puertos y enrutamiento"}, {"id":2,"nombre":"SSO","descripcion":"Keycloak, SAML, OpenID Connect, LDAP, autenticación y atributos"}]', annotations=None, meta=None)], structured_content={'result': [{"id": 4, "nombre": 'Balanceador', 'descripcion': 'HAProxy, Traefik, balanceo de carga, VIP, certificados y proxy inverso'}, {"id": 5, "nombre": 'Base de datos', 'descripcion': 'PostgreSQL, MariaDB, consultas SQL, conexiones y respaldos'}, {"id": 1, "nombre": 'HPC', 'descripcion': 'Clúster, Slurm, colas, nodos, GPU, software científico y almacenamiento HPC'}, {"id": 7, "nombre": 'Otro', 'descripcion': 'Solicitudes que no corresponden a los servicios anteriores'}, {"id": 3, "nombre": 'Plataforma web', 'descripcion': 'Drupal, WordPress, Joomla, Apache, PHP-FPM, Varnish y hospedaje web'}, {"id": 6, "nombre": 'Redes', 'descripcion': 'DNS, VPN, firewall, conectividad, puertos y enrutamiento'}, {"id": 2, "nombre": 'SSO', 'descripcion': 'Keycloak, SAML, OpenID Connect, LDAP, autenticación y atributos'}]}, meta={'fastmcp': {'wrap_result': True}}, data=[{"id": 4, "nombre": 'Balanceador', 'descripcion': 'HAProxy, Traefik, balanceo de carga, VIP, certificados y proxy inverso'}, {"id": 5, "nombre": 'Base de datos', 'descripcion': 'PostgreSQL, MariaDB, consultas SQL, conexiones y respaldos'}, {"id": 1, "nombre": 'HPC', 'descripcion': 'Clúster, Slurm, colas, nodos, GPU, software científico y almacenamiento HPC'}, {"id": 7, "nombre": 'Otro', 'descripcion': 'Solicitudes que no corresponden a los servicios anteriores'}, {"id": 3, "nombre": 'Plataforma web', 'descripcion': 'Drupal, WordPress, Joomla, Apache, PHP-FPM, Varnish y hospedaje web'}, {"id": 6, "nombre": 'Redes', 'descripcion': 'DNS, VPN, firewall, conectividad, puertos y enrutamiento'}, {"id": 2, "nombre": 'SSO', 'descripcion': 'Keycloak, SAML, OpenID Connect, LDAP, autenticación y atributos'}]}, is_error=False)
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter#
```

3.8 y 3.9

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python clasificar_tiquete_openrouter_mcp.py \
"El trabajo enviado a Slurm permanece pendiente por límite de QoS"
{
  "servicio_id": 1,
  "servicio": "HPC",
  "confianza": 0.9,
  "justificacion": "El ticket menciona Slurm, que es el gestor de trabajos de HPC."
}
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter#
```

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python clasificar_tiquete_openrouter_mcp.py \
"La aplicación no recibe el atributo ucrUserId después de autenticarse con Keycloak"
{
  "servicio_id": 2,
  "servicio": "SSO",
  "confianza": 0.9,
  "justificacion": "El ticket menciona Keycloak y la falta de un atributo de autenticación, lo que indica un problema con el servicio de SSO."
}
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter#
```

4.1 - 4.3

```
GNU nano 8.7.1 /etc/systemd/system/crm-mcp-openrouter.service
[Unit]
Description=Servidor FastMCP para clasificador de tickets con PostgreSQL
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
User=root
Group=root

WorkingDirectory=/root/clasificador_tiquetes_openrouter
EnvironmentFile=/root/clasificador_tiquetes_openrouter/.env

ExecStart=/root/clasificador_tiquetes_openrouter/.venv/bin/python /root/clasificador_tiquetes_openrouter/crm_mcp_server.py

Restart=on-failure
RestartSec=5

StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target

(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# sudo nano /etc/systemd/system/crm-mcp-openrouter.service
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# sudo systemctl daemon-reload
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# sudo systemctl enable --now crm-mcp-openrouter
Created symlink '/etc/systemd/system/multi-user.target.wants/crm-mcp-openrouter.service' → '/etc/systemd/system/crm-mcp-openrouter.service'.
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# sudo systemctl status crm-mcp-openrouter
● crm-mcp-openrouter.service - Servidor FastMCP para clasificador de tickets con PostgreSQL
   Loaded: loaded (/etc/systemd/system/crm-mcp-openrouter.service; enabled; preset: enabled)
   Active: activating (auto-restart) (Result: exit-code) since Wed 2026-07-22 22:15:56 CST; 2s ago
  Invocation: e4da172f704f426ca2c803a4d8a28076
    Process: 8509 ExecStart=/root/clasificador_tiquetes_openrouter/.venv/bin/python /root/clasificador_tiquetes_openrouter/crm_mcp_server.py (code=exited, status=3)
   Main PID: 8509 (code=exited, status=3)
    Mem peak: 98.5M
       CPU: 2.544s
Jul 22 22:15:56 SAULI systemd[1]: crm-mcp-openrouter.service: Failed with result 'exit-code'.
Jul 22 22:15:56 SAULI systemd[1]: crm-mcp-openrouter.service: Consumed 2.544s CPU time over 3.132s wall clock time, 98.5M memory peak.
lines 1-11/11 (END)
[1]+  Stopped                  sudo systemctl status crm-mcp-openrouter
```

4.4

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python clasificar_tiquete_openrouter_mcp.py \
"El sitio WordPress muestra error 500 después de actualizar PHP-FPM"
{
  "servicio_id": 3,
  "servicio": "Plataforma web",
  "confianza": 0.9,
  "justificacion": "Error 500 tras actualizar PHP-FPM afecta al sitio WordPress, que es una plataforma web."
}
```

5

Caso 1

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python clasificar_tiquete_openrouter_mcp.py \
"El trabajo de Slurm quedó pendiente por límite de QoS"
{
  "servicio_id": 1,
  "servicio": "HPC",
  "confianza": 0.9,
  "justificacion": "El ticket menciona Slurm, que es un gestor de trabajos de HPC."
}
```

Caso 2:

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python clasificar_tiquete_openrouter_mcp.py \
"No se envía el atributo email desde Keycloak al proveedor de servicio SAML"
{
  "servicio_id": 2,
  "servicio": "SSO",
  "confianza": 0.9,
  "justificacion": "El ticket menciona que el atributo email no se envía desde Keycloak al proveedor de servicio SAML, lo cual corresponde al servicio de SSO."
}
```

Caso 3:

```
(.venv) root@SAULI:~/clasificador_tiquetes_openrouter# python clasificar_tiquete_openrouter_mcp.py \
"El sitio Drupal devuelve 503 desde Varnish"
{
  "servicio_id": 3,
  "servicio": "Plataforma web",
  "confianza": 0.9,
  "justificacion": "El ticket menciona un sitio Drupal que devuelve 503 desde Varnish, lo cual corresponde a problemas de la plataforma web."
}
```